



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2025-26/SYST001/475681

Date:

11/11/2025(06:27:42)

Subject:LMI ON Prevention and Safe Clearing of Ash Build-up in Boiler
Bottom Ash Hoppers

Re-forwarded as instructed.

Kindly review and forward.

Regards

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LMI is in line with the revised OGN. Forwarded for further review.

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May pl review

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LOCAL MANAGEMENT INSTRUCTION

PREVENTION AND SAFE CLEARING OF ASH BUILD-UP IN BOTTOM ASH HOPPERS

LMI/OGN/OPS/SYST/001 Rev:10

MAY-2025

Approved for implementation by HOP (Simhadri)

LMI APPROVAL DOCUMENT

1	LMI NAME	Prevention & Safe Clearing Of Ash Build Up In Bottom Ash Hoppers
2	LMI No.	LMI/OGN/OPS/SYST/001 Rev:10
3	Reference Document	OGN-OPS-SYST-001-Rev07
4	Superseded Document	LMI/OGN/OPS/SYST/001, Issue: 01, Revision: 09, Apr'23
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TITLE	PREVENTION AND SAFE CLEARING OF ASH BUILD-UP IN BOTTOM ASH HOPPERS
LMI NO.	LMI/OGN/OPS/SYST/001 Rev:10

1.0 INTRODUCTION

This LMI is issued to comply with Operation Guidance Note COS-ISO-00 **OGN/OPS/SYST/001** Rev. No.: 10 MAY 2025. This guidance note identifies the need to exercise great care to avoid slag formation in the boiler furnace which results in poor heat transfer of furnace walls, large slabs of fused ash, blocking of clinker grinders, ash slurry pumps & choking of ash disposal lines. Excessive slag formation or improper bottom deashing leads to build-up of bottom ash to a height of 16-18 meters in the furnace bottom, forcing unit shutdown which could last several days. Build-up of this magnitude can also result in catastrophic failure of boiler support system. In line with above operation guidance note, this LMI will serve as a guidance instruction to avoid / minimize slag formation in the furnace of boiler & management of ash build ups in the bottom ash hoppers of NTPC Simhadri Stage I & II.

2.0 SUPERSEDED DOCUMENTS

LMI/OGN/OPS/SYST/001, Issue: 01, Revision: 09, Apr'23

3.0 SCOPE

LMI shall be applicable to all the four boilers & their associated bottom ash hoppers of Simhadri Stage I & II (4 x 500MW)

4.0 RECORDS OF REVISION

6.5 Isokinetic sampling to be done as per recommended schedule.

9.2. g. De-Nox burners should be operated in auto as per curve provided by OEM & while checking SADC, DE nox burners position also to be checked.

9.2.3 a Apart from AOH checking may be done during short shutdowns or opportunity shutdowns along with Boiler & C&I.

9.2.3 c. During AOHs, clearance for scaffolding removal inside boiler to be taken from Operation/ POG on combustion.

9.3.6 In water impounded BAH, Before commencement of deashing, BA hopper to be filled with water & overflow to be ensured.

9.4 peep holes above bottom ash hopper on both sides are to be regularly inspected physically for any sign of build up by Unit Controller

9.4.2 For ensuring complete emptiness of bottom ash hopper, deashing to be done at least once in a day in 'draw down mode'.

9.4.3 Once hopper is nearly empty, all hopper nozzles to be operated. Operation of hopper flushing nozzles will help in removing the remaining ash. If hopper nozzles are not provided appropriate modification to be carried out during AOH.

9.4.8 Cameras to be installed near 10.5 m near S-panel, 4.5 mtr at BAH view glass & near CG chamber view glass. Healthiness of camera to be monitored & status to be recorded in log book

9.4.9 After completion of bottom deashing, executive from control room shall physically ensure from local & record the same in log book.

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5.0 COAL-ASH SLAGGING

Slagging occurs when firing zone temperature goes above ash fusion temperature and fused ash either solidified on surfaces exposed to high radiant heat or fuse ash from clinkers and fall in boiler bottom hopper.

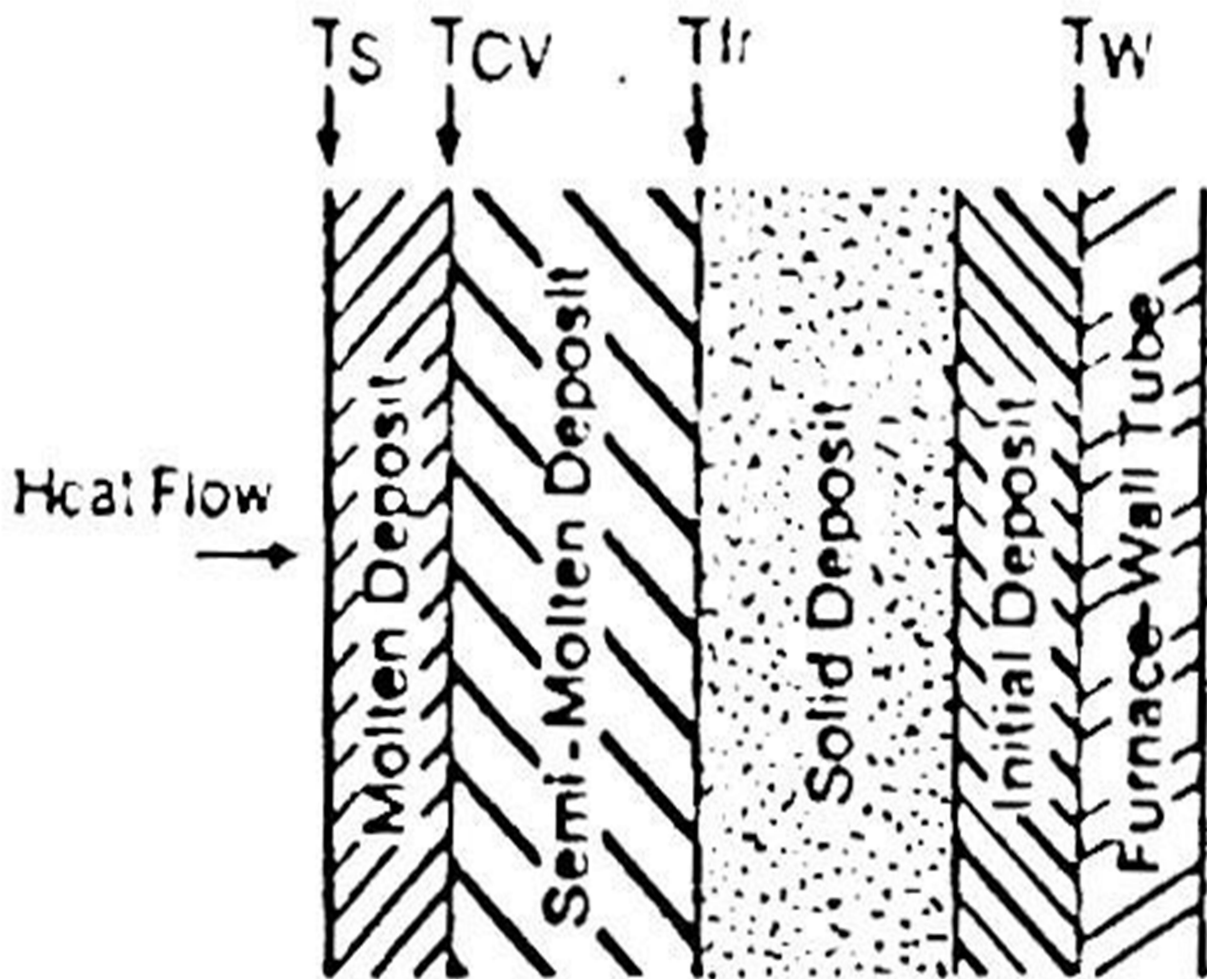
Coal parameters used for evaluating coal- ash behavior are:

- Ash-fusibility temperatures
- Base/acid ratio (More basic nature implies more tendency for slagging) One member for safety.
- Iron/calcium ratio
- Silica/alumina ratio
- Iron/dolomite ratio
- Dolomite percentage
- Ferric percentage

Molten to semi-molten coal-ash slag deposits will not form on clean water-wall tubes since, upon approaching relatively cooler tube surface, the slag particles become less adhesive because of rapid cooling in the wall-adjacent area. Accordingly, coal-ash deposition is a two- stage process. A primary layer of deposit first forms on the water wall tube surface. The resulting rise in the surrounding surface temperatures subsequently allows the adherence of rigid plastic secondary deposits. The figure below shows a typical cross section of a water wall slag deposit that has reached a maximum steady-state thickness.

Figure 1. Cross section of a typical furnace-wall slag deposit.

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TS: Slag Deposit Surface Temperature
TCV: Temperature of Critical Viscosity
Tfr: Freezing Temperature
TW: Tube Surface Temperature

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5.1 Mechanism of Coal-Ash Slag Deposition:

- The following two types of primary deposits are most commonly observed.
- Primary deposits, that results from the settling of the finer fractions (smaller than 30 microns) of fly ash. This type of primary deposit is loose in structure and does not provide cohesive or adhesive bonds.
- Primary deposits, that results from the selective deposition of certain reactive components of the ash (Iron, Calcium, or Alkalis). These components can be present in the deposit in high concentrations as oxides and/or Sulphur compounds, leading to the formation of low-melting eutectic mixtures. This type of deposit is denser in Structure and has stronger cohesive and adhesive bonds.
- During the deposition, there is a transitional stage where the plastic secondary deposits begin to stick on the primary layer. Time and increasing temperatures strengthen these secondary deposit/droplets of low viscosity molten glass. The temperature at which the secondary deposits begin to form corresponds to approximately 950-degree C. The transition from primary to secondary deposit is a function of the immediate gas temperature outside the deposit.

5.2 Deposit Structure and Maximum Thickness

The figure 1 showing the cross-section of a typical water wall slag deposit has a slag surface that has reached a maximum steady-state thickness. The rate at which glass droplets fuse together is directly dependent upon the time available for bonding to develop, and is inversely proportional to the viscosity of the melt, i.e., the more fluid the droplets are the more readily they will run together thus strengthening the mechanical bonding in the deposits, and ultimately thick slabs may built up, reducing the heat transfer efficiency of the boiler surfaces.

6.0 CONTROL OF SLAGGING

This section discusses the operational techniques that the operator can use to control slagging. Although the predominant factors affecting ash deposition are the amount and composition of coal ash; boiler operating conditions have also been demonstrated to affect deposition. The operating variables that influence slagging are:

- a) Furnace temperature
- b) Excess air
- c) Fuel fineness
- d) Secondary air distribution
- e) Distribution of coal

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- f) Air fuel ratio
- g) Mill combination in service
- h) Type of coal

Changing any one, however, will not eliminate operating problems completely. Although the individual variable affects boiler performance, they must be studied collectively to determine the effectiveness of a given programme. Effect of each operating variable is considered as follows:-

6.1 Furnace temperature:

Higher the furnace temperature on account of improper combustion, poor heat absorption in the walls or sometimes due to higher load, greater the potential for slagging. Changes in excess air, fuel fineness, and secondary air distribution are less drastic methods for minimizing slagging.

6.2 Excess air:

In a reducing atmosphere, the ash fusion temperature reduces. Here the Fe^{2+} ions will increase at the expense of more highly oxidized Fe^{3+} ions. The ferrous ions act as a flux and this reduces the initial deformation temperature of ash. If slagging is a problem with high iron coals, furnace deposits can be reduced drastically by increasing the amount of excess air. As a rule-of-thumb, the higher the fusion temperature, the drier the slag in the furnace, and the easier it is to remove.

6.3 Fuel fineness:

Slagging conditions often can be improved by control of the pulveriser fineness. Since coarse particles take longer to burn, hence these burn away from the intended zone of combustion. They are more prone to producing slag. High retention on +50 mesh often increases slagging tendencies. This is better accomplished by maintaining air/fuel ratio at designed level. Pulverized coal samples from all mills to be analyzed at least once a week to ensure the prescribed fineness levels. Any deviation needs to be addressed immediately.

6.4 Secondary air distribution:

Basically, the objective is to provide a good mixing of fuel and air so that combustion is efficient, and local zones with reducing atmosphere are avoided. In units with tangential firing, for example, slagging can be reduced by optimizing secondary air flow to fuel compartments. It is accomplished by regulating fuel air and combustion air distribution. The extent of Fuel air damper opening can be decided depending upon the amount of volatile matter in coal. In changing coal profile, such dampers have to be readjusted to get efficient combustion. Fuel air dampers of mills not in service (especially bottom mills) should be kept closed as they cause an increase in furnace temperature.

6.5 Fuel distribution:

Improper distribution of coal, due to eroded coal burner nozzles/ nozzle tips/ impellers will affect the combustion and hence increase the furnace temperature. Mill clean air flow test and Dirty pitot test to be conducted at regular frequency to assess the unbalance in pulverized fuel distribution and to take necessary corrective action.

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Isokinetic sampling to be done as per recommended schedule.

6.6 Other Operational Measures:

- Failure to remove deposits at the proper time may result in chain reaction of deteriorating events. For example, excessive furnace slagging can result from not using the wall blowers at proper intervals. This condition imposes higher gas temperatures in the convective section of the furnace because of the reduced rate of heat absorption. The attemperation levels indicate the level of deposits in a boiler for a constant steam temperature. High attemperation flow denotes a high deposit on the boiler tubes. Thus it is best to do soot blowing whenever such a situation arises.
- Using programmed soot blower systems, ash deposits on the furnace walls can be minimised. Waiting too long between operations can seriously hamper the effectiveness of soot blowers. Observation of furnace conditions at least twice per shift helps eliminate unexpected problems. Because the soot blower system is so important for reliable, full capacity operation of a coal-fired unit, this equipment must be maintained in good operating condition.

After overhauling of boiler / maintenance of furnace, the boiler should be inspected and loose material if any, is to be removed from furnace / bottom ash hopper. Falling of loose materials in the bottom ash hopper will obstruct ash flow inside the hopper / full closing of feed gates / jamming of clinker grinder / increase the time of deashing, etc.

7.0 INTERMITTENT REMOVAL - BOTTOM ASH HOPPER

Operating experience with water-filled hoppers has shown that most of the ash will flow towards and through the feeder gates by gravity; however, large slabs of fused ash do occur that will not move by gravity alone. Their removal requires the assistance of high pressure (8 to 9 kg/cm²) water jetting nozzles located on the sloping floors of the hopper. In addition, jetting nozzles in the wall opposite each feeder gate (wherever available) breaks-up the arching and packing of the ash that may occur at the gate. The principal functions/reasons for water-impounding include the following:

- To break-up large pieces of slag by thermal shock as they fall into the pool.
- To keep the ash and slag submerged so that they do not fuse into large unmanageable masses that would result if they were exposed to furnace heat.
- To provide a resilient medium to decelerate the large pieces of slag.

It is important to realise that the ash falling from the furnace walls does not pass through the furnace bottom opening in equal increments along the length of the opening. About 25% of the total ash to the furnace bottom will go through a 3 to 4 feet portion near the side wall and less than 2% of the bottom ash will go through each remaining foot of the bottom slot.

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8.0 REASONS FOR ASH BUILD-UP

8.1 Bottom Ash hoppers are designed to carry the ash for about six hours (with intermittent deashing system). However the carrying capacity in terms of time depends on various factors like coal firing rate, ash content in coal, bottom ash/fly ash ratio etc. While firing the lowest-heating value and highest-ash-content coal, the normal practice of emptying ash hoppers is once every 3 to 4 hours. The usual time required to evacuate the ash is 1 to 1 ½ hours. If deashing is delayed due to any reason, it must be attended immediately and bottom deashing started to avoid further ash accumulation.

8.2 Falling slag / hard clinkers also can cause accumulation inside the bottom ash hopper as they will block the hopper gates and prevent free flow of ash. These may lead to incomplete deashing also as the clinkers remain undetected inside the hopper.

8.3 Both incomplete and delayed deashing will cause the ash level to come above of water, which in turn converts to hard clinkers on exposure to heat from the furnace. This will in turn form a platform for further ash build up and if it crosses the S panel area it will be very difficult to dislodge it.

8.4 Care to be taken to ensure that refractory lining inside hoppers is smooth and there are no pockets (refractory damage) in the lining. These pockets will serve as a seat for subsequent ash build up in the hopper. Patching of refractory may preferably be done using Castable refractory (with 60% alumina content).

8.5 The ash build-up can be of two types from ash removing strategy perspective:

A. Continuous Deashing system with scrapper conveyors:

- i) Build-up during normal unit operation with bottom ash gates wide open.
- ii) Build-up during / after scrapper conveyor maintenance activities with bottom ash gates in closed condition.

B. Intermittent Deashing system with clinker grinders:

- i) Improper / Incomplete Bottom deAshing
- ii) Delayed Bottom deAshing

9.0 MEANS OF PREVENTING AND CLEARING ASH BUILD-UP

9.1 ACTIONS TO BE TAKEN FOR PREVENTING CLINKERING DURING COAL FIRING

9.1.1 Flue gas exit temperature on both sides at platen SH inlet to be analyzed and safe operating value of this parameter to be established for each unit. This parameter shall be denoted as FGET (PSH). It is observed in general high FGET (PSH) is indicating furnace leading to clinkering tendency. UCE of concerned unit should study the trend of this parameter and establish normal operating value of this FGET (PSH). An alarm should be provided above the normal operating value of this FGET (PSH).

Responsibility: AGM (EEMG) In consultation with-OS/UCE

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9.1.2 Mill Combination, SADC, Air flow and O₂ to be adjusted to achieve FGET (PSH) normal operating value.

Responsibility: UCE

9.1.3 In case of clinkering avoid running the unit with continuous mills in service. Split mill combination will help to reduce furnace temperature. Operate mills with 70 % (i.e. 45tph feed rate) of their rated capacity to reduce the heat load per burner.

Responsibility: UCE

9.1.4 SADC should be operated in auto mode. Wind box DP of 100 mm of WCL and as per the ramp given by OEM to be followed strictly. In case of clinkering DP bias may be varied to a maximum of ± 20 mm of WCL to check the improvement in clinkering parameters. It is to be normalized after observation if no improvement is noted.

Responsibility: UCE

9.1.5 SADC local scales are to be provided to easily understand the exact opening of SADC. SADC checking to be done in the shift in case of clinkering tendency noticed in the boiler. Corrective action to be taken immediately.

Responsibility: AGM-C&I/AGM (BMD)/AGM (O)

9.1.6 Keep all Mills PA flows as per curve implemented in auto. PA header pressure to be maintained as per operating value.

Responsibility: UCE/SCE

9.1.7 Always try to establish approximate coal flow through the feeder by monitoring the mill DP and Mill inlet temperature. Mill inlet temperature of mills with same amount of loading needs to be approximately same. If not, need to check for feeder calibration.

Responsibility: UCE and AGM-C&I

9.1.8 Feeder calibration is to be done every six months

Responsibility: MTP, C&I and SCE

9.1.9 In clinkering conditions it is highly essential to maintain oxidizing atmosphere in boilers. Maintain O₂ level $\geq 3.5\%$ (design).

Responsibility: UCE

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9.1.10 Water wall soot blowing is to be done every shift during clinkering considering unit flame stability. Attention to be given for reduction in spray levels and also furnace temperature during wall blowing.

Responsibility: UCE

9.1.11 Dirty Pitot tube test to be done, if the deviation in coal flow among the four corners is more than 10% ,mill should not be taken in service.

Responsibility: EEMG&BMD

9.1.12 Bottom Ash Sample physical monitoring to be done.

Responsibility: UCE/ Shift chemist

9.1.13 Burner tilt is to be operated uniformly in all corners. Burner tilt should be checked locally for equal operation on all corners and all elevations. Delinking of some of the burners from outside is one of the reasons for clinkering.

Responsibility: UCE/BMD/C&I

9.1.14 The following mill parameters are to be monitored

SL.NO	Parameter	Operating Value	Limit value
1	Coal flow	16.8 -60 TPH	63/68 TPH FOR STG-1/2
2	Air flow	Min Value:70TPH Max value:95 TPH	102 TPH
3	Mill DP	150 mmWCL	300 mmWCL
4	Mill Outlet Temp	60-80deg	90 deg
5	Mill I/L temp	250-300 deg	325deg

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9.1.15 S/H & R/H spray, furnace draft variation, load variation with same coal feeding and clinker grinder currents to be monitored. It is to be ensured that CG current is available in OWS. A spike in the clinker grinder current indicates the hard clinkers. LRSB may be operated to dislodge any clinkers especially during clinkering.

Responsibility: UCE/EMD/C&I

9.1.16 CHP to inform unit control room regarding type of coal bunkering done. All steps to be taken to prevent bunkering high quality of coal in bunkers.

Responsibility: SCE-CHP

9.1.17 Furnace temperature measurement to be done with portable pyrometer and normal operating values for different mill combinations to be established for each unit.

Responsibility: HOD (EEMG)

9.1.18 Pulveriser coal pipes metal temperature monitoring to be done twice in each shift to identify the coal pipes choking. It is to be ensured that coal pipe temp points are available in OWS.

Responsibility: Boiler Desk engineer/ C&I

9.1.19 Clean air flow test and Dirty Pitot tube test to be done and test as per CENPEEP specified procedure and frequency. The results to be made available at unit control room. Fuel fineness to be measured and reports should be updated.

Responsibility: HOD (EEMG) &HOD (chemistry)

9.1.20 During short shutdowns, all coal elevation buckets position free operation to be checked through manholes/peepholes using powerful torch light, search light. This will identify any bucket that has fallen and always remaining in -30 degrees. If any such dropping of bucket found, immediately maintenance to rectify the same.

Responsibility: AGM (BMD)/AGM (O)

9.1.21 Calibration of PA flow transmitters to be done in each overhaul and PA inlet temperature to be done once every year.

Responsibility: C&I

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9.2 POINTS TO BE CHECKED DURING OVERHAULS TO PREVENT COMBUSTION PROBLEMS

9.2.1 SADC CHECKING

- a) All SADC to be checked for mechanical freeness.
- b) All SADC are provided with stopper to maintain minimum opening of SADC even in full close condition.
- c) All SADC to be connected with Power cylinders from C&I and check each and every individual damper opening as per control room command.
- d) All SADC are provided with Indicators indicating position of SADC to the nearest 5 percent.
- e) Over fire dampers are to be operational from UCR and they should be available for operation.
- f) Bottom SADC (below AA) should be kept at 30 to 50 percent.
- g) De-Nox burners should be operated in auto as per curve provided by OEM & while checking SADC , DE nox burners position also to be checked .

Responsibility: BMD/OPRN/EEMG and C&I

9.2.2 COAL BURNERS CHECKING

- a) Identify the mills whose coal nozzles were damaged to the maximum extent and try to correlate them with mill operating conditions by monitoring historical data.
- b) Once the unit comes into service these mills clean air flow test and dirty Pitot tube tests are to be done compulsorily and establish the reason for damage.
- c) All coal nozzles are to be replaced with new ones. Reclaimed coal nozzles not to be used if two years' overhauls strategy is followed.
- d) Mill orifices to be checked during every overhaul.

Responsibility: BMD/OPRN/EEMG

9.2.3 BURNER TILT CHECKING

- a) All coal nozzles burner tilt operation to be checked and a combined protocol to be signed by operation and Boiler maintenance.
All coal nozzles operation to be checked and angle measurement to be done with protractor as a check with operation. Apart from AOH checking may be done during short shutdowns or opportunity shutdowns along with Boiler & C&I.
- b) Power cylinder connected and operation to be checked from inside and outside as well along with operation, C&I and Boiler Maintenance. Combined protocol in this regard is to be signed by all concerned groups.
- c) During AOHs , clearance for scaffolding removal inside boiler to be taken from Operation/ POG on combustion

Responsibility: BMD/OPRN/C&I

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9.3 BEFORE DEASHING:

9.3.1 On getting indications of slagging or high furnace temperature, the combustion related parameters (Burner Tilt, SAD/FAD, and Secondary Air Flow) are to be checked and necessary corrective actions taken accordingly.

Responsibility: UCE

9.3.2 To prevent build up in bottom ash hopper and to start deashing in time, peep holes above bottom ash hopper (at 10.5m) on both sides are to be regularly inspected for any sign of build up by Unit Controller. This is to be recorded in the logbook. This observation is to be made preferably three hours after completion of dashing or whenever slagging /ash build up tendency exists. It is to be ensured that BAH & BAH overflow lines temperatures are displayed in OWS. This will help for early identification of ash buildup.

Responsibility: UCE/Boiler Operator/C&I

9.3.3 Thermocouples, cameras at strategic locations like bottom ash hoppers, need to be installed and hooked up to control room which will help to analyse and prevent ash build up as early as possible.

Responsibility: AGM-C&I

9.3.4 On detecting any sign of ash build up, bottom deashing is to be commenced immediately. Due to any reason if it cannot be started immediately, load on the unit should be reduced and ash level shall be monitored continuously.

Responsibility: UCE/SCE

9.3.5 In the normal course frequency of bottom deashing is to be decided based on the coal firing rate and hopper filling/ash build up trend.

Responsibility: UCE/SCE

9.3.6 Before commencement of deashing, BA hopper to be filled with water & overflow to be ensured.

Responsibility: UCE/SCE

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9.4 DURING DEASHING

9.4.1 During bottom deashing it is to be ensured that the feed gates are opened fully for free flow of bottom ash. Drains of the Ash Disposal lines should be opened after starting bottom deashing to check for proper discharge. Adequate discharge from lines to be ensured at Ash slurry sump. A pale colour of discharge indicates slow evacuation. The ejector water pressure is important as a lower pressure will make evacuation slower. After starting bottom deashing the side view glasses of the bottom ash hopper need to be monitored for level drop. If the view glasses are getting cleared on one side only, it indicates a build-up/improper deashing on the other side. Should there be any problem in the discharge, back flushing of the ejector can be done to normalize the discharge. Ash build up on the side of the hopper (Left or right) is to be tackled by opening the side flushing valves.

Responsibility: UCE

9.4.2 Deashing of bottom ash hopper is to be carried out in 'maintained level mode'. This would result in additional suction head for jet pump and also eliminate problem of dry ash as ash will always be quenched. In order to check for any ash build up in the bottom ash hopper, deashing may be done occasionally (say, once in a day) in 'draw down mode' for a short duration.

Responsibility: UCE

9.4.3 During bottom deashing, loading on clinker grinders is to be observed from the ammeters provided in AHP control room / local panels/ Main Plant OWS. In case ammeters are not provided, they should be installed in both AHP & main plant. Observations are also to be made at ash slurry falling in the ash slurry pit for sufficient quantity of ash in slurry reaching there. Bottom ash discharge line pressure also indicates the quality of discharge (More for proper discharge and vice versa).

Responsibility: UCE

9.4.4 Entire bottom ash is evacuated from the bottom ash hopper to be ensured and there is no build up of ash over S-panel of furnace. Any abnormality must be reported to SCE and Head of Operation.

Responsibility: UCE

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9.4.5 When the hopper becomes empty observation is to be made from the view glasses for deposition of bottom ash, if any, in the hopper. This will also be facilitated by cameras installed for the same. In case there is any leftover bottom ash, the other set of diagonally opposite clinker grinders are to be taken in service after stopping the two clinker grinders which have been in service. In case ammeters are not provided, they should be installed in both AHP & main plant. Operation of hopper flushing nozzles will also help in removing the remaining ash. After ensuring that bottom ash hoppers are empty, observation is made through peep holes provided above S-panel of furnace to look for any ash build up there. Wherever peepholes are available above the grinders, visual inspection of the hopper is to be done to ensure an empty hopper. A typical empty hopper demonstrates fast moving ash pieces.

Responsibility: UCE

9.4.6 The inspection view glasses/peepholes are easily accessible and a clear view maintained. Availability of adequate provision for cleaning of View Glass

Responsibility: UCE/ HOD (AHM)

9.4.7 The bottom ash hoppers are not to be left dry for long durations to avoid ash build up on the slopes of the hopper.

Responsibility: UCE

9.4.8 Cameras to be installed near 10.5 m near S -panel, 4.5 mtr at BAH view glass & near CG chamber view glass. Healthiness of camera to be monitored & status to be recorded in log book.

Responsibility: UCE

9.4.9 After completion of bottom deashing, executive from control room shall physically ensure from local & record the same in log book.

Responsibility: UCE

9.5 IN CASE OF ASH BUILDUP

9.5.1 Unit load shall be brought down to technical minimum once any ash accumulation is observed through bottom most furnace peep hole. Drum pressure to be maintained as recommended to avoid reduction in heat absorption at water wall leading to high furnace flue gas exit temperature. All efforts must be made to remove ash buildup in running condition. Lower most Oil Gun support to be taken to stabilize the flame. Poking or use of water lance shall be avoided for dislodging the ash in unit running condition.

Responsibility: UCE/SCE

9.5.2 If all actions fail to remove/dislodge the massive slag deposits, in two shifts or the ash accumulation goes above any of the S-panel peep holes (around 10 m level) /flame is not visible through peephole (one or both side), whichever is earlier the unit should be taken under shut down for removal of clinkers/ slag accumulated in the furnace. Taking a short shut down to clear the ash build up is desirable than prolonged operation under reduced load in view of loss of generation.

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Accumulated ash and clinkers may get further fused and solidified making it risky and time consuming to clear it with passing of every hour. Immediately after unit shut down bottom ash hopper is to be flooded with water to avoid hard clinkers formation from burning slags. In no case unit is to be run with uncontrolled and unmanageable slag deposits or without bottom ash evacuation.

Responsibility: SCE/HOD (O)

9.5.3 Manholes at level above the bottom ash hopper shall not be opened while the boiler is in service. Boiler to be shut down and cooled before opening manholes.

Responsibility: HOD (BMD-PP)

9.5.4 All safety procedures shall be adhered before starting of the job of opening the manholes and clearing the ash build up inside the bottom ash hopper. PTW must be taken and a committee comprising of Head of Operation, BMD, Ash handling and Safety should inspect the area and declare area safe for work on a protocol. Same committee shall inspect the area again after clearing of ash from the bottom ash hopper and give clearance before the start of other jobs.

Responsibility: HOD (BMD-PP), HOD (AHM) &HOD (Safety)

9.5.5 Head of (O&M) shall ensure that all works around the boiler area including bottom ash hopper area be suspended, and people working in the area be removed, from the time of observation of Ash buildup till complete clearing of ash.

Responsibility: HOD (O&M)

9.5.6 Concerned persons involved in ash / slag / clinker removing activities should stand at safe distance from bottom ash hoppers so that risk of splashing ash / hot water causing injuries is minimised. Water lances should have sufficient long metallic pipes so that cleaning is done from a safe distance. A practice of carrying out one job only at a time, i.e., either clearing of bottom ash hoppers or clearing of slag formation above S-panel is to be adhered to.

Responsibility: HOD (BMD-PP), HOD (AHM) &HOD (Safety)

9.5.7 Personal Protective Equipment (PPE) is to be used by personnel, supervising and executing the job of deashing. The PPE includes safety helmets with face shield, gum boots, fire retardant hand gloves and fire-resistant overall suit. All Personal Protective Equipment (PPE) are to be kept in common place like shift maintenance room in a separate box for easy availability during the emergency of choking removal for saving time.

Responsibility: HOD (BMD-PP), HOD (AHM) &HOD (Safety)

9.5.8 All efforts should be made to avoid the practice of by-passing the clinker grinders and dumping the slurry on boiler floor.

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Responsibility: SCE/HOD (Operation)

9.5.9 While shutting down Unit, sufficient care needs to be taken to avoid overheating of boiler tubes. Boiler is not to be drained and drum level is to be maintained.

Responsibility: UCE

9.5.10 Manhole door near S-panel may be opened after unit shutdowns to clear the ash build up. Care to be taken to provide wire mesh cover near the manhole door of S-panel to minimize the chance of personnel getting exposed to hot ash/gases from inside the furnace.

Responsibility: HOD (BMD-PP)

9.5.11 OTHER PRECAUTIONARY MEASURES

- a) Since feed gates are very critical items their reliability must be ensured by carrying out thorough maintenance during overhauls.
- b) Routine changeover of clinker grinders shall be carried out regularly as per the schedule.
- c) In case any of the critical equipment like feed gates, clinker grinders, hydro ejector is under break down it must be attended at the earliest, if required by taking unit shut down because effect of keeping it in such condition can be far reaching.
- d) Facility of quick filling of bottom ash hopper (within 10 – 15 minutes), wherever not provided, should be provided.
- e) Dry run of Clinker Grinders (feed gates open and no water in the Bottom Ash Hopper) shall be avoided. Operation of LRSB during bottom deashing shall not be done.

Responsibility-HOD (O)/HOD (AHM)/HOD (C&I)

9.5.12 PRECAUTION DURING IMPORT COAL FIRING

It is observed that slagging tendency increases while firing imported blended coal. As ash fusion temp of imported coal is less as compared to Indian coal. Slagging tendency of coal is dependent on various composition ratio like Iron calcium ratio, silica alumina ratio & base acid ratio. However, due precaution to be taken to avoid ash build up while firing imported coal. Details regarding the same is attached in annexure as SOP on import coal firing.

Responsibility-HOD (O)/HOD (BMD)/ HOD (AHM)/HOD (CHP)/ HOD (C&I)

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10.0 CONCERNED LOGS FOR ASH BUILD UP PREVENTION

- A) UCE log book
- B) Report format for ash build up-Annex-1.
- C) Protocol for opening s-panel/bottom ash hopper manhole after unit shutdown- Annex-2.
- D) Check list for “safe clearing of bottom ash buildup”-Annex-3
- E) SOP on Import coal firing- Annexure-4

11.0 REVIEW

AGM (O) shall arrange for review and take the approval of competent authority. The document will be reviewed 2 yearly basis or as and when required.

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CHECK LIST FOR SAFE CLEARING OF BOTTOM ASH BUILDUP

SL.NO	DESCRIPTION	REMARKS
1	Committee comprising of Head of Operation, BMD, Ash handling and Safety to be formed.	
2	Entire area to be cordoned to avoid un-necessary movement around BAH area.	
3	PTW to be taken after committee clears the area is safe to work.	
4	Boiler to be allowed to cool before opening of manhole. No LRSB (1st pass) or wall blower to be operated to depressurise the boiler .	
5	All safety procedures shall be adhered before starting of the job of opening the manholes and clearing the ash build up inside the bottom ash hopper.	
6	Personal Protective Equipment (PPE) are to be used by personnel, supervising, and executing the job of deashing. The PPE includes Thermal wear, safety helmets with face shield, gum boots, fire retardant hand gloves and fire resistant overall suit.	
7	All Personal Protective Equipment (PPE) are to be kept in common place like shift maintenance room in a separate box for easy availability during the emergency of chockage removal for saving time.	
8	Ensure concerned persons involved in ash / slag / clinker removing activities should stand at safe distance from bottom ash hoppers so that risk of splashing ash / hot water causing injuries is minimised.	
9	BAH to be cleared first followed by ash above S-panel	
10	Boiler bottom Seal trough sealing is to be maintained	
11	Bottom ash gate opening to be regulated if SC/CG to avoid overloading of scrapper conveyor or clinker Grinder.	

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12	Wire mesh cover near the manhole door of S-panel to be provided to minimize the chance of personnel getting exposed to hot ash/gases from inside the furnace.	
13	CG changeover to be ensured & disposal of slurry at sump to be checked.	
14	Quick filling of bottom ash hoper to be ensured.	
15	Post cleaning of ash build up ,boiler & BAH internal inspection to be done before final box up.	

Signature			
Name			
Designation			
Department	AHD	BMD	OPRN

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ANNEXURE:1 Report Format for Ash Build Up

S.NO	Description	A shift	B shift	C shift
1	Blending done in bunkers			
2	SADC /BT checking			
3	10.5m Peep holes			
4	Bottom deashing duration			
5	Clinker Grinder not available			
6	Unburnt in BA & FA			
7	Wall blowing			
8	LRSB blowing			
9	Max furnace temp			
10	All running coal pipe temp			
11	Bunker temp			
12	Any hot spot in bunker conical portion			
13	Fire in Mill rejects			
14	Mills not available			
15	Mill i/l PA-mix temperature			
16	PAPH/SAPH air o/l temp			
17	FG temp at Chimney			
18	FEGT			
19	Max PSH metal temp			
20	Max RH metal temp			

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Annexure – 2: SOP on Import coal firing

Precautions & Parameters to be monitored while firing blended coal

- Co-ordinate & communicate with CHP regarding bunkering of imported coal at shift beginning and shift end & to be recorded.
- While blending Efforts to be made to keep the specific coal between 0.6 and 0.8.
- Sp. coal of unit & bunkers in which imported coal is blended to be recorded.
- Coal pipe temp of the mills to be monitored.
- Mill parameters like mill DP, current & outlet temp of mills.
- Mill reject system & pyrite hopper to be checked once in every hour.
- Mill temp after stopping the mill also to be observed.
- Ensure feed rate is not less than 50%.
- Furnace temp to be monitored hourly if furnace temp probes are not available temp gun may be used.
- 10.5 m peep holes to be checked every hour
- SH/RH spray & flue gas exit temp to be recorded.
- Furnace draft to be under constant watch for any abnormal fluctuations.
- Unburnt in fly ash & BA to be measured in every shift.
- Flue gas O₂ (L/R), furnace to wind box DP & total air flow.
- SADC opening at local
- Ash sample to be checked for stickiness during every deashing.
- As fire coal sample to be collected from individual feeders & to be analyzed on feeder basis to ascertain blending ratio & should be informed to CHP. This check to be done on weekly basis.

Work instruction:

- O₂ in flue gas to be maintained >3.5 % at full load & >5.5 % at technical minimum load.
- Split Mill combination to be preferred over continues mills.
- Furnace temp. mapping to be done & recorded regularly during the shifts.

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High range temp guns to be procured on urgent basis & to be kept available in control room.

- FEGT to be restricted to 900-950 deg C.
- Over fire dampers to be operationalized & to be kept in auto
- SADC checking to be done during every shift & deviations to be attended
- LRSB & wall blowing schedule to be followed.
 - In addition to the above, increase in SH/RH spray without change infiring regime indicates Fouling in upper furnace, hence wall blowing is to be done. Decrease in SH/RH spray without change in firing regime indicates fouling in SH /RH path, hence LRSB to be done.
- SH/RH spray flow to be recorded, Mill changeover may be done between the available mills. Burner tilt may be operated as per limiting values of metal temp.
- Any fluctuation in furnace pressure may be an indication of spontaneous combustion in Mills.
- Bunkering of blended coal to be done uniformly in all bunkers.
- Furnace to wind box DP to be maintained as per OEM guidelines.
- Firing during Ramp up/down should be controlled in such a way that there is gradual increase in spray. Sudden excessive increase in spray to be avoided.
- All tube metal temperature in FG path & penthouse to be hooked up in PI.
- Alarm to be provided for change in sp. Coal below 0.6. Alarm for rate of change of sp. coal (beyond ramp period) may be explored.
- Any aberration or difficulty in following SOP to be discussed with COS committee on Imported coal firing

Bottom ash Area

- While firing blended coal, de-ashing schedule to be followed & ensure no delay.
- Bottom deashing duration to be monitored, if any delay in emptying the chamber or BAH is observed it should be investigated.
- BAH to be filled & overflowing to be ensured all the time. BAH overflow line temp. to be observed
- Unburnt in Fly ash & bottom ash sample to be checked
- While doing deashing, confirm with AHP that slurry discharge is adequate. if

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required back flushing to be done.

- Cross Clinker grinder are preferably to be taken current to be monitored & grinders to be changed during every deashing. Any deviation in Deashing time to be investigated.
- BAH view glass level should come down after 15 to 20 min of start deashing & visibility of view glass should be adequate.
- Check 10.5 m peep holes before & after deashing.
- During deashing all hopper flushing valves to be operated to avoid any build up in corners
- If despite doing above if BAH is not getting emptied, grinders may be changed & tried again.
- If ash build has gone up Bottom ring header /S-panel peep holes & ash evacuation is not taking place, shutdown of unit to be taken.

Mills

Precautions

- Mill outlet temp to be maintained as per the guideline given however thumb rule for mill outlet temp set point should be (100 -VM in coal).
- Ensure inert steam system is charged condition & fire hydrant valve & hose should be available.
- Ensure no accumulation of coal dust and PF leakages from MDV area, Coal pipe couplings, Burner floor etc.
- In case of any PF leakage mill should be stopped immediately and any PF accumulation should be cleared immediately.
- No additional biasing for mill PA flow
- Monitor each mill coal pipe temperatures less than 70 deg C with temp. gun. Stop the mill and inform maintenance if temperature is maintaining more
- Mill fineness to be in range of 70-75 %.
- Mill dirty pitot tube test may be done for the mills where blended coal will be fed.
- Mill reject system to be monitored for smoke, fire, chock

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- Lean air flow mixture to be avoided
- If fire splinters are observed from many Mills, then MRHS may be stopped & reject system may be kept opened with target plates in front of reject mouth. Adequate safety to be taken to avoid spread of splinters.
- Mill inerting system to be revived & made available.
- Mills to be rotated on regular basis. No mill to be kept out of service for more one week.
- If ash build has gone up Bottom ring header /S-panel peep holes & ash evacuation is not taking place, shutdown of unit to be taken. (Please refer OD on Prevention & clearing of ash build up in Bottom ash hopper)

Guideline while Starting of Mill

- Establish ignition support in boiler before starting the mill.
- Mill inerting system is charged up to common isolation valve
- Start the mill & let reject come out of reject system.
- Maintain air flow of 50 TPH with CAD for at least 5 min.
- Open HAG/HAD & maintain the mill outlet temp as per the curve.
- Feeder loading should be maintained above 50 %.

Guidelines while Stopping of Mill

- Slowly bring the federate to minimum
- Maintain mill outlet temp within the prescribed limits.
- Before stopping the feeder mill inlet temp to be reduced to 100 deg c.
- After stopping the feeder, cold air (70 tph) to be given for minimum 5 min.
- Mill to be run without any air for further 5 min.

Emergency stopping of mill in case of fire

- In case of sudden increase in mill outlet temp or coal pipe temp, mill should be tripped immediately.
- Mill BIGs /MDVs to be closed.
- Charge inerting steam till temp start coming down
- Allow the mill to cool to ambient temp.
- Do not start the mill to evacuate coal from mill.

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- No PTW to be issued till mill is completely cooled.

Starting of Mill post fire

- Charge inerting steam before starting of Mill.
- Physical inspection of Milling system from Mill to BIG (burner isolation gate) to be checked for any abnormality
- Keep mill reject mouth in full open condition.
- After clearance from mill maintenance & keep the mill in service & empty the coal
- Purge the coal pipes for 30 min with cold air. (40 tph) ensuring ignition support.

Bunker

- Always be in touch CHP control & record the type of coal fed & physical level of bunker in logbook.
- Regularly Check for any smoke or fire at bunker floor.
- Regularly check the temp of bunkers using temp gun.
- Check for any hotspots on conical portion of bunker.
- In case of fire in bunker, SOP on bunker fire & Feeder fire to be followed.
- Ensure fire water booster line to all TPs are in charged condition.

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ANNEXURE-3 GUIDELINES FOR PROPER BLENDING OF COAL

For making a homogenous mixture of imported and domestic coal the following blending guidelines are to be followed:

1. Imported coal is first to be stacked in identified yard at earmarked stockpiles.
2. Mixing of imported coal with domestic coal in track hopper is to be avoided.
3. GCV of imported coal lying in different stockpiles is to be measured at regular intervals and record is to be made available in CHP Control Room.
4. Imported coal is to be drawn from stockyard through stacker reclaimer / underground hopper / over ground hopper and is to be mixed with domestic coal (being conveyed from track hopper/ wagon tippler) at the merging transfer point in a pre-defined ratio. Further homogenization for a more uniform product takes place in upstream transfer points.
5. Under no circumstances Import coal should be directly sent to bunker.
6. In case of high GCV coal received from domestic sources such as (ECL, NEC etc where GCV is more than 4500), no imported coal blending is to be done.
7. Belt weighers of evacuation conveyor stream from track hopper / wagon tippler and stacker reclaimer machines are to be kept operational with rated accuracy and good communication between control rooms, paddle feeder operator, stacker reclaimer operator is to be established for maintaining blending ratio.
8. Each coal station should frame 'station specific coal blending guideline' giving equipment/ conveyors (shown on CHP layout), stockyards/stockpiles to be taken in service for the purpose of blending
9. GCV of coal received from high GCV domestic coal sources to be analysed in express lab, so that GCV is known to CHP control room within 2 hrs.
10. Details of incoming coal rake should be noted by CHP to know source of coal (i.e. company/ mines / grades).
 - From Indian Railway FOIS site
 - From RR
 - From Slip pasted on coal rake by knowing the source of the coal
11. CHP control room to keep a data base of GCV of various mines, so that by knowing the mine's name, Control room can guess about the GVC of the coal.
12. If there is no feeding /flow from domestic coal stream, then after 2 minutes import stream should be tripped.
13. CHP Control Room engineers to be sensitized of detrimental effect of rich blending on boiler like mill/boiler explosion.

TITLE	PREVENTION AND SAFE CLEARING OF ASH BUILD-UP IN BOTTOM ASH HOPPERS
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Annexure-4 Exception Report Format for Imported coal firing

Station Name:

Date:

S.NO	Description	Remarks
1	Bunkering done in all bunkers or not. If not any problem may be specified	
2	FEGT Excursion in any units.	
3	10.5m Peep holes clear in all units	
4	Specific coal going below 0.55 (give details)	
5	Clinker Formation tendency in any units	
6	Unburnt in BA & FA within limits	
7	Wall blowing	
8	LRSB blowing	
9	All running coal pipe temperatures are ok or not.	
10	Any hot spot in bunker conical portion	
11	MTM Excursion if any (unit wise)	
12	Any other observation	

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